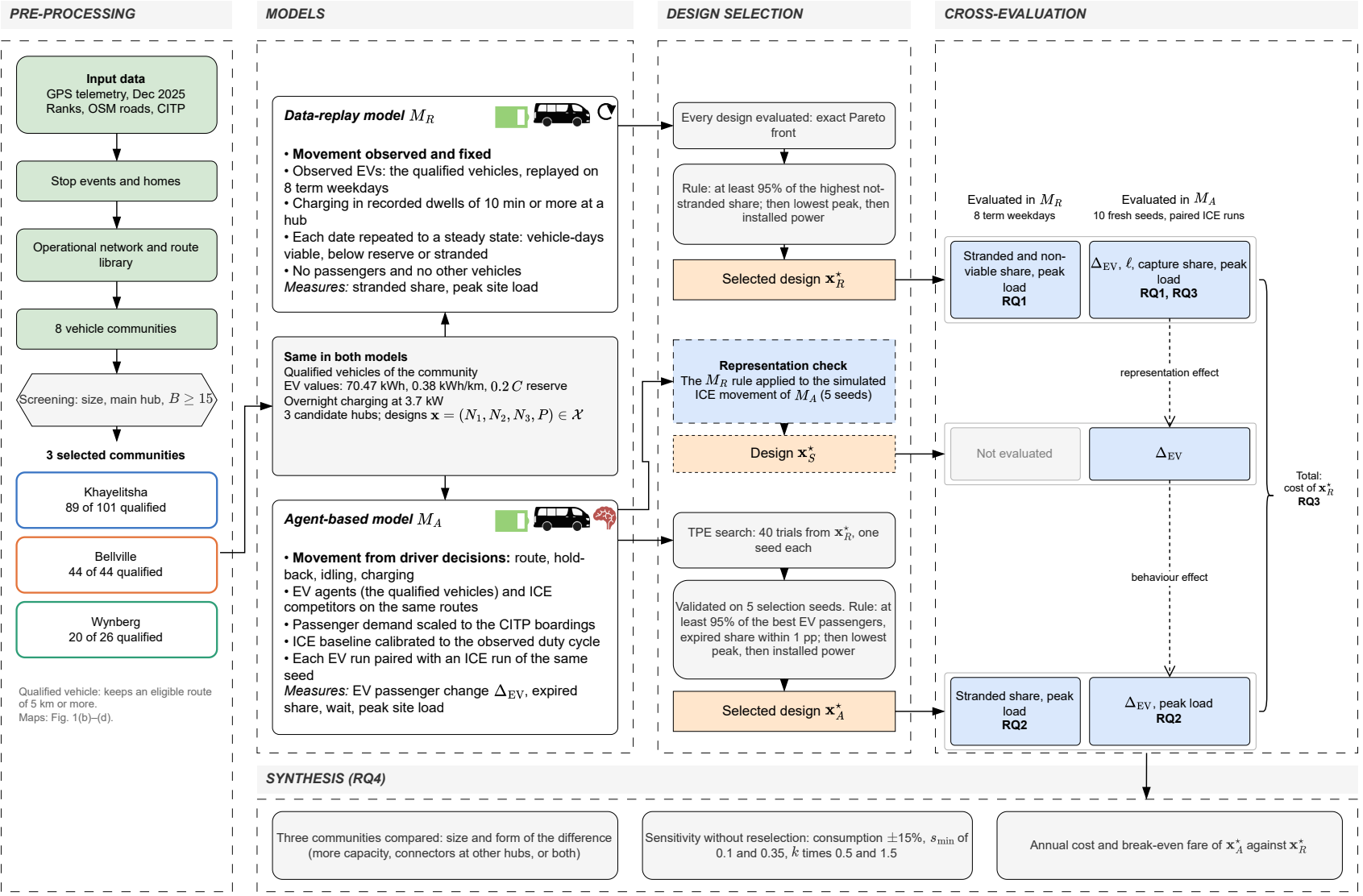

PRE-PROCESSING
MODELS
DESIGN SELECTION
CROSS-EVALUATION
Input data
GPS telemetry, Dec 2025
Ranks, OSM roads, CITP
Stop events and homes
Operational network and route library
8 vehicle communities
Screening: size, main hub, $B \geq 15$
3 selected communities
Khayelitsha
89 of 101 qualified
Bellville
44 of 44 qualified
Wynberg
20 of 26 qualified
Qualified vehicle: keeps an eligible route of 5 km or more.
Maps: Fig. 1(b)–(d).
Data-replay model M_R
• Movement observed and fixed
• Observed EVs: the qualified vehicles, replayed on 8 term weekdays
• Charging in recorded dwells of 10 min or more at a hub
• Each date repeated to a steady state: vehicle-days viable, below reserve or stranded
• No passengers and no other vehicles
Measures: stranded share, peak site load
Same in both models
Qualified vehicles of the community
EV values: 70.47 kWh, 0.38 kWh/km, 0.2 C reserve
Overnight charging at 3.7 kW
3 candidate hubs; designs $\mathbf{x} = (N_1, N_2, N_3, P) \in \mathcal{X}$
Agent-based model M_A
• Movement from driver decisions: route, hold-back, idling, charging
• EV agents (the qualified vehicles) and ICE competitors on the same routes
• Passenger demand scaled to the CITP boardings
• ICE baseline calibrated to the observed duty cycle
• Each EV run paired with an ICE run of the same seed
Measures: EV passenger change Δ_{EV}, expired share, wait, peak site load
Every design evaluated: exact Pareto front
Rule: at least 95% of the highest not-stranded share; then lowest peak, then installed power
Selected design $\mathbf{x}_R^\star$
Representation check
The M_R rule applied to the simulated ICE movement of M_A (5 seeds)
Design $\mathbf{x}_S^\star$
TPE search: 40 trials from $\mathbf{x}_R^\star$, one seed each
Validated on 5 selection seeds. Rule: at least 95% of the best EV passengers, expired share within 1 pp; then lowest peak, then installed power
Selected design $\mathbf{x}_A^\star$
Evaluated in M_R
8 term weekdays
Evaluated in M_A
10 fresh seeds, paired ICE runs
Stranded and non-viable share, peak load
RQ1
Δ_{EV}, ℓ, capture share, peak load
RQ1, RQ3
representation effect
Not evaluated
Δ_{EV}
behaviour effect
Stranded share, peak load
RQ2
Δ_{EV}, peak load
RQ2
Total: cost of $\mathbf{x}_R^\star$
RQ3
SYNTHESIS (RQ4)
Three communities compared: size and form of the difference (more capacity, connectors at other hubs, or both)
Sensitivity without reselection: consumption ±15%, $s_{\min}$ of 0.1 and 0.35, k times 0.5 and 1.5
Annual cost and break-even fare of $\mathbf{x}_A^\star$ against $\mathbf{x}_R^\star$